

Scalability & Future Plan

Date	02 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited crop dataset and soil data coverage	May reduce prediction accuracy for different regions	High
2	Supports only web-based access	Mobile users cannot access the system easily	Medium
3	No real-time weather data integration	Recommendations may not reflect current weather conditions	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users	Deploy on cloud platforms (AWS/Azure/Render) with load balancing to support thousands of users.
2	Data Storage	Stores crop data in a local MySQL database	Use cloud databases with automated backup and scalable storage.
3	Performance	Predictions are generated on a single server	Optimize the ML model and use caching for faster response times.
4	Security	Basic user authentication	Implement role-based access, encrypted passwords, and HTTPS communication.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve crop recommendation accuracy using larger datasets	3 Months	More accurate predictions and better crop selection
Phase 2	Develop an Android/iOS mobile application	6 Months	Improved accessibility for farmers
Phase 3	Integrate real-time weather and IoT sensor data	9 Months	More accurate recommendations based on live environmental conditions
Phase 4	Add fertilizer, irrigation, and pest management recommendations	12 Months	Complete smart farming assistance and improved agricultural productivity